Health-Project 3: Breast Cancer Detection

Experiment 2: Dataset Details & Data Audit

Course: CSE4001 — Applications of Machine Learning in Industry | **Source:** UCI Machine Learning Repository

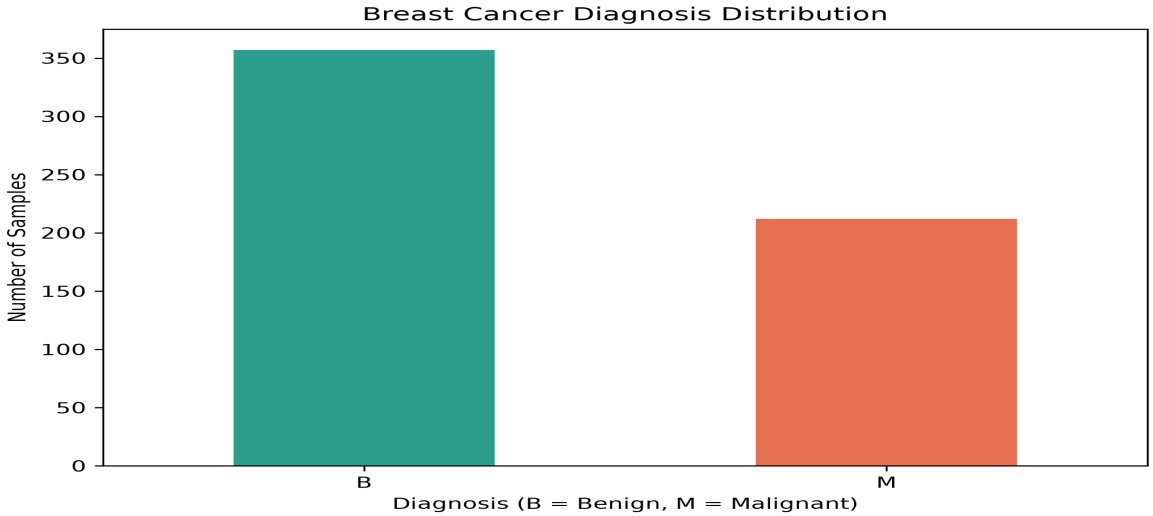
1. Dataset Source & Provenance

The Wisconsin Diagnostic Breast Cancer (WDBC) dataset was created by Dr. William H. Wolberg, W. Nick Street, and Olvi L. Mangasarian (1995) at the University of Wisconsin. It consists of 569 fine-needle aspirate (FNA) biopsy samples characterizing cell nuclei morphology digitized from microscope slides.

2. Dataset Dimensions & Schema

Attribute Group	Count	Details / Description
Total Records	569	Individual patient biopsy image extractions
Metadata Field	1 (id)	Unique patient sample identifier (non-predictive)
Target Variable	1 (diagnosis)	Ground truth: Benign (B) or Malignant (M)
Feature Measurements	10 base metrics	Radius, Texture, Perimeter, Area, Smoothness, Compactness, Concavity, Concave points, Symmetry, Fractal dimension
Total Predictive Features	30 continuous	Each of 10 metrics calculated as Mean, Standard Error (SE), and Worst (mean of 3 largest values)
Missing / Null Values	0 (0.0%)	Complete dataset verified across all 569 rows

3. Target Distribution & Class Balance



Target Distribution: Benign = 357 (62.74%), Malignant = 212 (37.26%). The moderate class imbalance confirms that accuracy alone is insufficient and requires precision-recall optimization.

4. Known Limitations & Data Quality Findings

- **Zero Missing Entries:** Verified 0 null values across all 32 raw attributes.

- **No Duplicate Biopsies:** Verified 569 unique records without replication.
- **Exclusion of ID Column:** The sample ID field carries no predictive biological signal and will be dropped in Experiment 3.
- **Cohort Scope:** Historical single-center dataset from 1995; multi-center external validation is advised before broad clinical deployment.